

Level blockout: a Team Deathmatch warehouse

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GitHub repository: github.com/bsse23040/ITU-CS-464-LAB-03-BSSE23040

Game I chose: PUBG Mobile, in its Team Deathmatch mode (the warehouse map).

For this lab I made a simple grey-box level in Unity from the provided POLYGON Prototype Pack by Synty. It has no scenery and no colour: only grey boxes, with one marked default route from the spawn to the goal. This document has my reference images, screenshots of the level, an explanation of the level and its obstacles, and the reasons for my design. The screen recording is a separate file, LevelWalkthrough.mp4, and it follows the default route.

1. Reference material

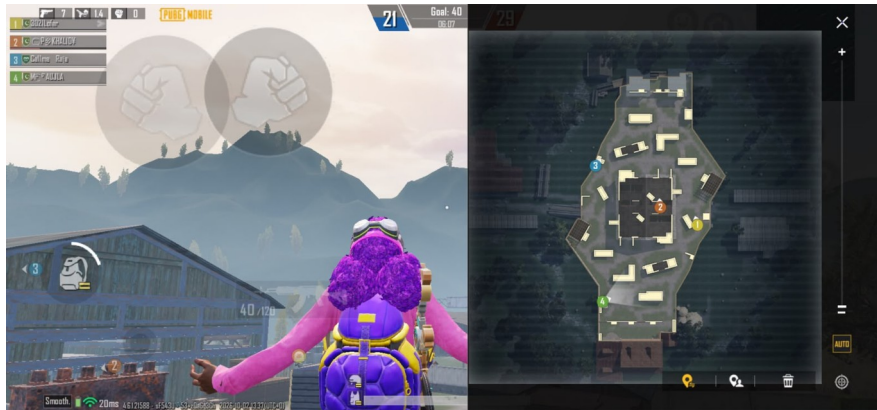
These are screenshots I took in a Team Deathmatch match, and I also recorded a short gameplay video (PUBG_gameplay_video.mp4, in the References folder of the repository). They are what I looked at while I built the level.



Reference 1: a big two-storey warehouse in the middle, with containers and crates around it.



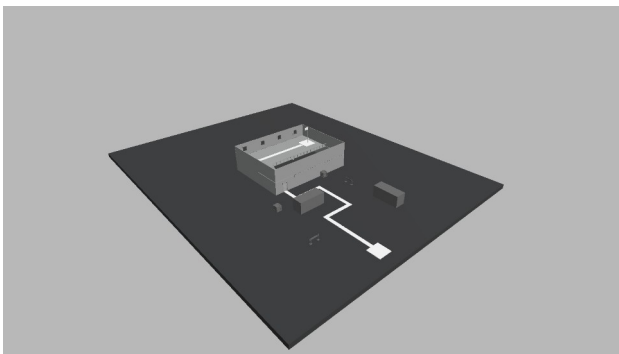
Reference 2: a lane between a wall of containers and wooden crates. An enemy is standing on a container roof.



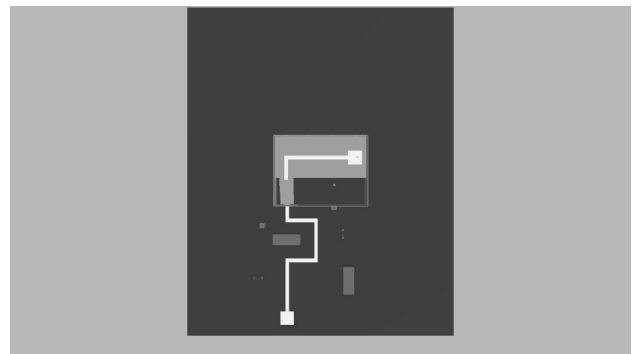
Reference 3: the full map. The arena is small, with a big central building and containers around it.

What I took from them: a big warehouse with two floors in the middle of the map, containers and crates as cover in the open ground in front of it, and a way up to the top floor, since the roof and upper floor are where the strong positions are.

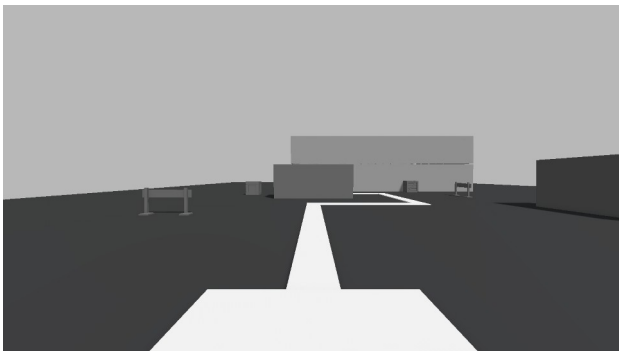
2. The level: Warehouse (Level_Warehouse.unity)



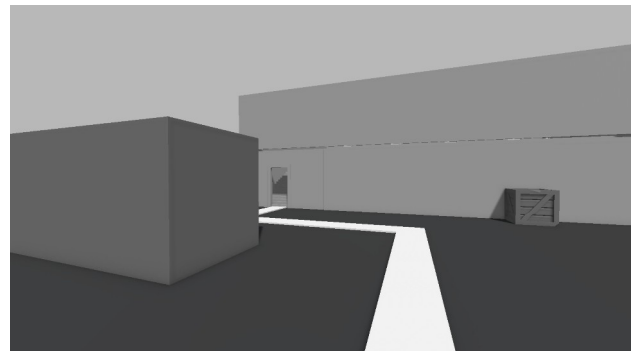
Overview of the whole level.



Top-down: the light strip is the default route.



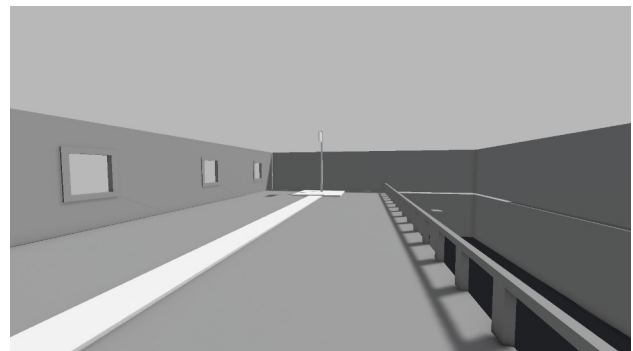
At the spawn, looking along the route.



The yard: the container forces a turn.



The doorway and the stairs behind it.



Upstairs: the last stretch to the goal.

What the level is

A flat grey ground with one two-storey warehouse (20 m by 15 m) and a small yard in front of it. The player starts on the white spawn pad at the south end and has to reach the white goal pad with the flag on the upper floor. Everything is grey: dark grey ground, mid grey walls and cover, and near-white only for the spawn, the goal and the route.

The default route

The route is marked on the floor with a light strip, so there is one clear way from the spawn to the goal. It is about 65 m long:

- **1.** Start on the spawn pad and walk straight north for 13 m.
- **2.** Turn east and walk 6.5 m, to get around the container that blocks the direct line.
- **3.** Turn north and walk 9 m past the side of the container, then turn west for 6.5 m.
- **4.** Go north to the doorway in the south wall of the warehouse.
- **5.** Climb the 3 m wide stairs (3.1 m high, 6 m long) straight ahead of the doorway.
- **6.** Walk north, then east along the upper floor for 15 m to the goal pad and flag.

Obstacles

- **The container in the yard.** A grey container (6 m by 2.6 m by 2.4 m), too big to see over or jump, sits right on the straight line from the spawn to the doorway. It forces the player to turn and take the long way round.
- **Cover.** A second container, two barriers and two crates stand in the open ground as extra cover, away from the route.
- **Doorway and stairs.** The only way into the warehouse is one doorway, and the only way up is the stairs right behind it, so the entrance is a chokepoint.
- **The warehouse itself.** The upper floor has window openings along the north wall and a rail along the stair opening, and two columns stand on the ground floor.

Why I designed it this way

- It follows my reference: a big two-storey warehouse in the middle, with containers and crates around it.
- I kept it grey and simple on purpose, so the shapes, the sizes and the route are what you see, not the art.
- One marked route means anyone opening the level knows where to go. The turns around the container make the route more than a straight walk, so it has a start, a challenge in the yard, a chokepoint at the door, a climb and a goal.
- The stairs are 3 m wide and climb about 27 degrees, the doorway is wider than the player and the walls are 3 m high, so the sizes feel right for a shooter.

3. How it was built

The lab asked for game objects and the provided assets, so I used both. The provided asset is the POLYGON Prototype Pack by Synty. I imported only the 13 pieces the level needs (about 1 MB) into Assets/Synty: the 5 m wall, doorway and window modules, the 5 m floor tile, the stairs, the rails, a column, two crates, a barrier, a flag pole and the block piece. The block piece, scaled, makes the ground, the containers, the spawn and goal pads and the route strip. I then gave every piece one of four plain greys, so there is no colour.

A small editor script (Assets/Editor/SimpleLevels.cs) places each piece by its measured size, so the walls and floors line up on a 5 m grid. The level is in Assets/Scenes/Lab03 in the repository. I used Unity's Scene view to look around, and the walkthrough video is a camera flown along the default route in the Scene view.

	Level_Warehouse
Size	60 m by 74 m ground, 20 m by 15 m warehouse

	Level_Warehouse
Default route	About 65 m, spawn to goal, marked with a light strip
Challenge	A container forces a detour; a doorway and stairs make the climb
Goal	White pad and flag on the upper floor
Colours	None: four greys only

The Scene view walkthrough is in LevelWalkthrough.mp4, and everything (scene, scripts, screenshots, references) is in the GitHub repository linked at the top.